MEDIUM · ADVANCED MATH

Advanced Math

10 questions · ~15 min

10

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

$$f(\theta) = -0.28(\theta - 27)^2 + 880$$

An engineer wanted to identify the best angle for a cooling fan in an engine in order to get the greatest airflow. The engineer discovered that the function above models the airflow $f(\theta)$, in cubic feet per minute, as a function of the angle of the fan θ, in degrees. According to the model, what angle, in degrees, gives the greatest airflow?

A. –0.28

B. 0.28

C. 27

D. 880

Q2 GRID-IN ADVANCED MATH

$$x^2 + x - 12 = 0$$

If a is a solution of the equation above and $a > 0$, what is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ADVANCED MATH

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

A. $-3k + 17r + 3$

B. $-3k + 17r - 3$

C. $-3k - 7r - 3$

D. $-3k - 7r + 3$

Q4 GRID-IN ADVANCED MATH

$$fx = x^5 + 9x + 17$$

For the given function f, the graph of $y = fx$ in the xy-plane passes through the point $0, b$, where b is a constant. What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 ADVANCED MATH

$$y = x + 1$$
$$y = x^2 + x$$

If (x, y) is a solution to the system of equations above, which of the following could be the value of x ?

A. –1

B. 0

C. 2

D. 3

Q6 ADVANCED MATH

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

A. $6x^2y^22x^6$

B. $6x^2y^2x^4$

C. $6x^2y^2x^6 + 2$

D. $6x^2y^2x^4 + 2$

Q7

ADVANCED MATH

For the polynomial function f, the graph of $y = fx$ in the xy-plane passes through the points -50, 10, and 40. Which of the following must be a factor of $f(x)$?

A. $x + 1$

B. $x + 4$

C. x - 1

D. x - 5

Q8

ADVANCED MATH

If $4\sqrt{2x} = 16$, what is the value of $6x$?

A. 24

B. 48

C. 72

D. 96

Q9

ADVANCED MATH

$$gx = \frac{3}{5}x + \frac{7}{6}$$

$$hx = 6x - 5$$

The functions g and h are defined by the equations shown. Which expression is equivalent to $gx \cdot hx$?

A. $\frac{18x^2}{5} - \frac{35}{6}$

B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$

C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$

D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

Q10 ADVANCED MATH

Immanuel purchased a certain rare coin on January 1. The function $f(x) = 651.03^x$, where $0 \leq x \leq 10$, gives the predicted value, in dollars, of the rare coin x years after Immanuel purchased it. What is the best interpretation of the statement "$f8$ is approximately equal to 82" in this context?

A. When the rare coin's predicted value is approximately 82 dollars, it is 8% greater than the predicted value, in dollars, on January 1 of the previous year.

B. When the rare coin's predicted value is approximately 82 dollars, it is 8 times the predicted value, in dollars, on January 1 of the previous year.

C. From the day Immanuel purchased the rare coin to 8 years after Immanuel purchased the coin, its predicted value increased by a total of approximately 82 dollars.

D. 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.